Topic 1.0

COMPUTER SYSTEM(S): Concepts and Definitions

LEARNING OUTCOMES

By the end of this topic, you should be able to:

1. Background of the course
2. Define what is meant by Computer systems
3. Identify types (standalone and networking) and roles of computer systems
4. Development of computer systems
5. Acquisition of computer systems
6. Evaluating and justifying computer systems investment
7. Identify the importance of computer systems

1.1 Background of Computer Systems

The term computer is/was originally derived from the word compute, which means calculate. The innovation behind computer was to perform superb calculations than the normal calculator. More specifically, the idea was to solve complex arithmetic and logical operations as well as scientific problems. In this contemporary world, computers have gone beyond the initial objectives with superb performances such as accepting data and information, storage, processing, sharing, security intelligence (SI), and artificial intelligence (AI). Any input by a user into the computer systems such as figure/text, voice, photo or video is called data. The outcome or output of the generated data by the computer systems is called information. Computers were also called data processors because of the capability of converting data into required information. This approach is/was called data processing, thus, data processor. The professionals' approach (instruction) that controls computer systems to perform the various tasks is known as computer programming.

In this Topic, you will be also exposed to the definition of computer systems, types and roles of computer systems in our societies and organisations, evolution of computer systems, and the importance. Before the details of those mentioned above, let us now refresh our memory to the initial set-up of computer systems/IT/ICT/computing/digital technology in our country.

The focus of computer systems development in Ghana includes:

i. Enhancing Ghana position as global digital and multimedia hub
ii. Expanding the communication network to ensure more equitable access to information and services
iii. Intensifying efforts to bridge the digital divide (digital literacy)
iv. Developing existing cybercities as well as promoting new cybercenters and multimedia applications

Apart from those focuses, measures will also be taken to enhance computer systems-related skills and competencies. Effort will also be made to improve broadband connections so as to allow usage of advanced multimedia applications, increase local content development, and enhanced e-commerce adoption.

1.2 Definition of computer systems

Computer systems refer to different units connected together to perform a specific task. Also, computer systems refer to integration of hardware, software, and other related components to work together to meet organisations and users' needs. Thus, computer systems consist of different units or components with various or logical roles such as keyboard, mouse, and system unit. More specifically, computer systems are made up of a various integrated units or components, which enable the entire structures to work effectively. On the other hand, each unit cannot function independently for the entire systems to work. See the Figure below for some units or components of computer systems.

Computer is an electronic device which performs high speed arithmetic and makes a logical decision under the control of instructions supplied by human. It is automatic and stores data and information for future use.

Systems (system) refer to a collection of different units or elements connected together to achieve a particular objective. It is a group of integrated parts in digital technology arranged together to perform a specific task to meet organisations and users' needs. In computing, systems are made up of more than one unit or element. The various elements work together with different role according to each unit.

Computer systems can also be defined as the study of computers' processes and structures to aid routing activities in human life. Computer systems provide users with the ability to use computational thinking and creativity to understand and solve routine activities such as at homes, offices, and schools. Thus, computing is the practice of using computers to solve a problem. In terse, computing is the process of calculating something with computers. Or it is the study of computers.

In terse, computer system(s) can be defined as exploring the complex terminologies and concepts in digital technologies.

1.3 Types and roles of computer systems

Computer systems consist of two (2) types, namely; standalone and networking.

1.3.1 In the context of standalone

Standalone is any computer system with all the necessary peripherals that can process users' routine activities without any network support. In this context, the system unit is the main component, which cannot be replaced by any part for the entire systems to work. Each system unit with the associated required components is able to work effectively and independently to meet organisations and users' needs without any support from other servers or terminals. The standalone computer systems are usually own by individuals and small organisations. The Figure below is an example of standalone computer systems.

In terse, information technology or information systems professionals or experts should note that computer systems or architecture focuses on three key areas, namely; processors, memories, and I/O components with their associated software drivers, or just software.

First, *processors* area is the key aspect. This section concentrates on computation (e.g., arithmetic) and control (e.g., executing a sequence of steps). This aspect is explained in other chapters more on contemporary central processing unit (CPU).

Second, the *memories* area is always considered next in computer systems. This section focuses on the storage (main or primary memory) and access of digital data/information.

Finally, *I/O components or peripherals* area is considered last in computer systems. This section focuses on the interconnection and standardisation of computers and devices such as display units, keyboards, mice, microphones, and disks.

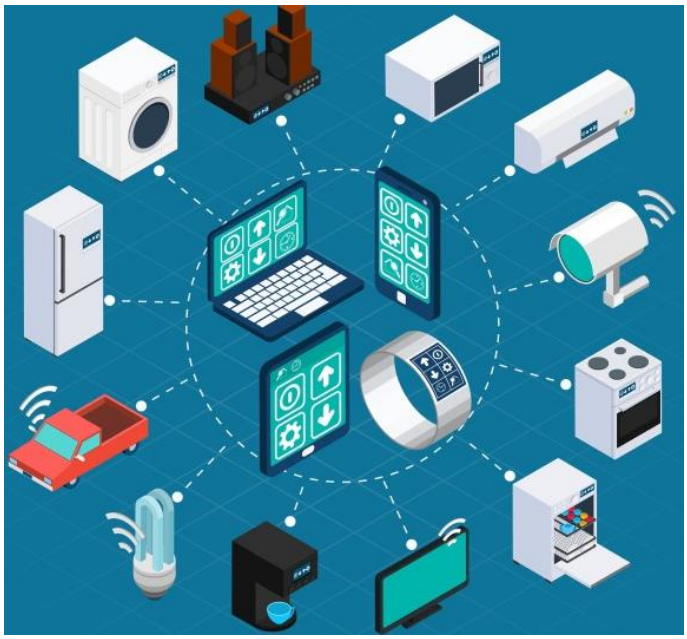
The organisations and users' applications associated with standalone computer systems are across different sectors for routine activities. The table below highlights some applications of standalone computer systems.

Some Applications of Standalone Computer System(s)

Sector	**Application**
Education	Lecturers standardizing students' results, preparing their teaching materials, and other related activities.
	Students preparing their assignments, project works, and other related activities
Church /other organisations	Record keeping of daily activities, record keeping of members' profiles, payment transactions, and other related activities
Copy center	Printing of customers' files, editing of customers' files, scanning of files for customers, and other related activities
Retail / Sales	Malls/supermarkets - Sales transactions, payment transactions, discount transactions, reorder transactions, and other related activities

1.3.2 In the context of networking or integrated computer systems

Computer networking is the other type of computer systems. This type consists of several standalone computer systems or terminals with a central controller or system called server. This category of computer systems connects several computers or digital devices across different locations or geographical areas with various users simultaneously to share data and information for their routine activities. The networking computer systems, integrated computer systems or computer networking systems are usually own by small and large organisations. The Figure below is an example of networking computer systems.



The applications of computer networking systems or integrated computer systems are pervasive across every digital-oriented organisation and society for real-time routine activities. Understanding computer systems enables you to know the applications and interactions among all digital related devices. The table below provides some highlights of computer networking systems applications in both societies and organisations.

Some Applications of Computer Networking Systems

Sector	Application
Education	Universities with integrated tertiary management systems for managing admissions, payments, programme/course registrations, publication of exams results, graduations, and other related activities.
	Lecturers providing assignments, providing feedbacks or responses to students' assignments, submitting students' results for publications, providing teaching services via online, and other related activities
	Students checking their results online, discussing and solving assignments via online platform. submitting their assignments via online, participating in e-leaning, and other related activities.
	Research- computer networking systems support researchers in the academic field to share data and information to solve a particular research problem

	virtually irrespective of each participant's location. This approach also enables researchers to solve complex problems with different methodologies within a shorter period.
Church/other organisations	Spreading the Good News to their members via digital platforms, providing announcements, payment transactions and acknowledgments, and other related activities
Health	Integrated health management systems adopted by the health sector facilitate patient record capturing and keeping, patient profile management, sharing of patients' profiles across different health sectors, bills payment transactions associated with funds providers, and other related activities.
	Health records associated with Ghana card, which provide an effective revenue mobilization to stop leakages in patient bill payments.
Transport	Transport - ridesharing platforms use integrated digital systems (networking) to link riders and drivers to share information for trip activities, integrated systems for trip payment via online, and other related activities.
Banking	Salary- digital-oriented organisations use computer networking systems associated with financial and other related organisations for salary payment transactions, contribution transactions, and other related activities.
Retail / Sales	Malls or supermarkets use computer networking systems to link other systems, various outlet or branches together to manage inventory or reorder, customer relationship, discounts transactions, and prices, digital payment transactions, and other related activities
Security	Ability to monitor and control vehicles, doors, and other digital-oriented items virtually. These applications minimize or prevent the risks likely to affect your assets.

1.4 Development of computer systems

The development of computer systems has gone through several waves or generations. Each generation has its own strengths and weaknesses. Despite the different generations of computers as at this contemporary world, new innovations are coming in day and day out. These numerous advancements in computing make experts in the field difficult to where we have reach. There are several authors with different categories of computer generations. The table below provides abstraction of generations of computers.

Generation	Major Development	Example
0 (before 1940)	Mechanical computers	Babbage Analytic Machine
1 (1940-1950)	Electronic computers by using vacuum tubes	ENIAC, EDVAC
2 (1950-1960)	Solid state electronics	IBM 7904
3 (1960-1970)	Small and medium scale integrated circuit	IBM System 360, DEC PDP-8
4 (1970-2000)	Large scale integrated circuit Development of microprocessors	IBM System 370
5 (2000- now)	Very Large-Scale Integrated Circuit, more powerful microprocessors, Semiconductor memory, smart processor-based or chips-based. This technology focuses on advancement operations such as multipurpose processing, cloud computing, simultaneous processing, anthropomorphic processing, ChatGPT, robotics, and artificial intelligence (AI).	PC, smart phones, drones. IBM's Watson

1.5 Computer systems architecture and infrastructure

It is important for us to differentiate between two important terms: computer systems architecture and ICT infrastructure.

1.5.1 The Concept of Computer Systems Architecture

The term “architecture” (or physical architecture) is commonly used to describe the way people's current and emerging requirements are combined with knowledge of the surrounding environment, resource availability, various constraints, and the technologies available to provide a blue print for construction of a building.

The concept of ICT architecture specifically in the context of networking is similar to the building’s architecture. ICT architecture serves as a broad blue print for acquiring, developing, implementing, and integrating ICT in order to support the business functions and strategies of an organisation. It integrates the information requirements of the organisation and all users, the ICT infrastructure and all ICT applications of the organisation.

In designing a physical architecture there are several layers of abstraction that need to be distinguished. This is done to maintain an overview of the complexity and to maintain an overview of the connections. This zooming in on different levels also happens with ICT architecture. We might for example talk about ICT architecture at enterprise level, ICT architecture in a project for several ICT applications and the architecture of an individual ICT application.

1.5.2 Documenting Computer Systems Architecture

ICT architecture of an organisation should be properly documented. There are many ways to describe, present, and document computer systems architecture, specifically in the context of networking. For example, different perspectives can focus on the following:

i. Services and processes to be supported.
ii. Clusters of functionalities required.
iii. Information and data that must be recorded and exchanged.
iv. Specific technology products deployed and integrated.
v. Technical standards for building systems and for interoperability between systems.

1.5.3 ICT/Computing Infrastructure

The main purpose of any computing infrastructure is to support and manage all ICT resources and services in an organisation. Thus, an organisation's ICT infrastructure consists of three elements as shown in Figure 1.4.

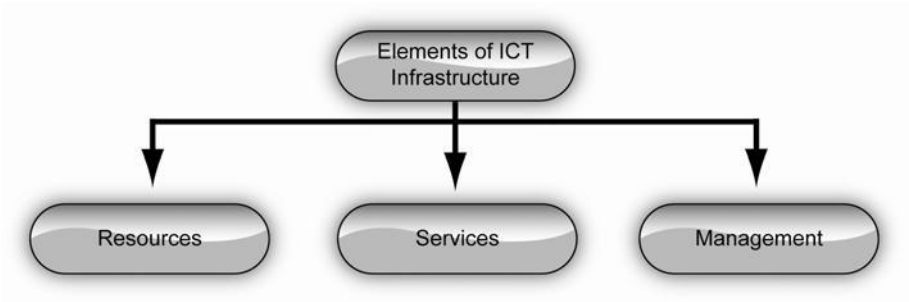


Figure 1.4: Elements of Computer Systems Infrastructure

Computer systems or ICT resources include:

i. Computer hardware, which is the physical equipment used for input, processing, storing and output activities;
ii. Computer software. The first type of computer software is called the system software, which consists of computer programs that control the computer

hardware. The second type of computer software is the application software;

iii. Communication Technology, consisting of both physical devices and software, links various computers located at various locations; and

iv. Data.

ICT services provide means for utilising all ICT resources available within the organisation. Examples of ICT services are electronic learning (e-learning), e-commerce, and knowledge management. *ICT management* explains how the available ICT resources and ICT services are arranged, operated, maintained, and managed.

1.5.4 Preparing Computer Systems / ICT Architecture

The process of preparing computer systems architecture in the context of networking must start with ICT planning. The ICT planning process starts by getting a holistic perspective on what the organisation aims to achieve and how it will do so. The process involves the steps shown in Figure 1.5 below.

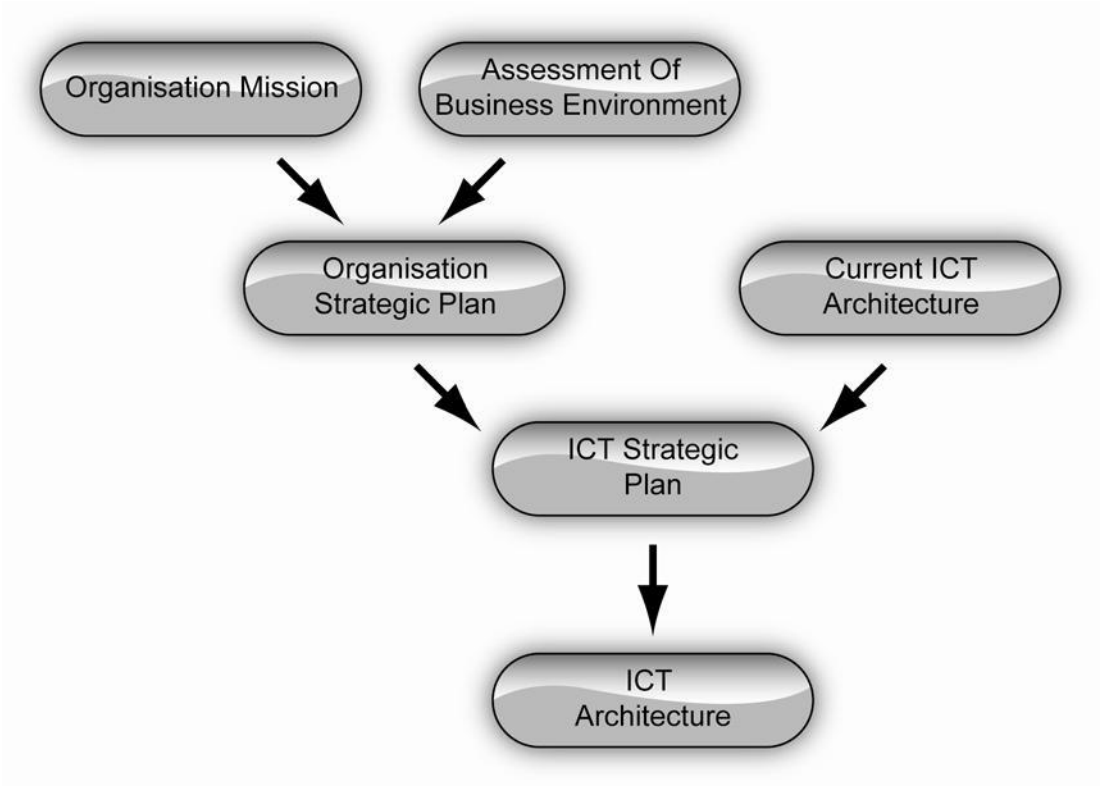


Figure 1.5: Developing ICT Architecture

The process of preparing ICT architecture involves analysing the organisation's strategic plan and the current ICT architecture, if it is already available. Based on this analysis, ICT Strategic Plan is developed. ICT Strategic Plan is a set of long-range goals that describes the ICT infrastructure needed to achieve the organisation's goals. A new ICT architecture is then developed based on the ICT strategic plan of the organisation.

The ICT Strategic Plan must be aligned to the overall organisation's strategic plan, so that the ICT department and other organisation's units are working towards the same goals. Achieving this alignment is a difficult and most organisations continue to demonstrate a limited actual alignment.

1.6 Computer Systems /ICT Acquisition

The acquisition is the process of obtaining computer systems infrastructure. The acquisition issue is complex for various reasons, such as:

i. There is a large variety of computer hardware, communication networks, and IT applications available; and
ii. These computer components such as hardware, communication networks, and IT applications keep changing over time.

1.6.1 Computer system(s)/ ICT Acquisition Process

The acquisition process must be based on the organisation's computer systems or ICT architecture. The process involves the steps as shown in Figure 1.6.

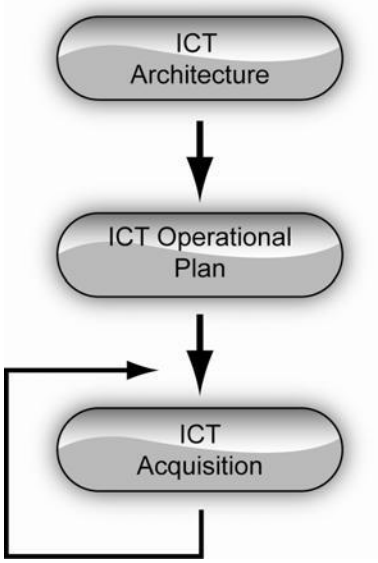


Figure 1.6: ICT Acquisition Process

The following strategies are important for the acquisition of ICT resources:

i. Analyse and understand the organisation's ICT architecture,
ii. Develop ICT Operational Plan, and
iii. Acquisition process.

First, the analyse and understand the organisation's ICT architecture give a broader scope of the type of ICT resources that can suite a particular organisation's needs.

Second, the ICT Operational Plan is a clear set of ICT projects that will be carried out by the organisation to support ICT Strategic Plan. This plan consists of the following elements:

i. Mission statement about the roles and functions of ICT in the organisation
ii. The organisation's ICT environment
iii. Objectives of the ICT functions
iv. Constraints of the ICT functions
v. Long term ICT infrastructure requirement
vi. Short term ICT infrastructure requirement ICT acquisition projects to be implemented during the current year.

Finally, the acquisition process is the approach to purchase the required ICT resources to meet what the organisation needs. This stage is the transaction for exchange money with ICT resources between the organisation setting up the ICT infrastructure and the one providing the products.

1.7 Evaluating and justifying computer system(s) investment

Justifying computer systems investment is done by using cost-benefit analysis. This analysis involves activities as described in Figure 1.7.

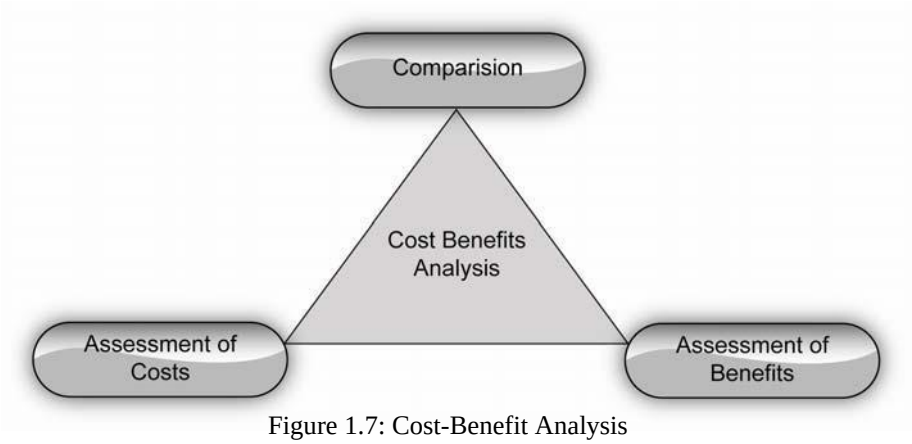


Figure 1.7: Cost-Benefit Analysis

One of the major issues is how to determine the cost of computer systems or IT investment. The normal approach is to use the concept called total cost of investment. By using this concept, the cost of computer systems investment is calculated by adding together the following:

i. The acquisition cost for hardware and software
ii. The operation or maintenance cost
iii. The training cost
iv. The control cost, such as the cost of standardisation and security control.

There are a number of values that can be used to measure benefits:

i. Return of Capital Investment
ii. Improvement in efficiency of providing a service
iii. Improvement of customers and partners' relationship
iv. Improvement of decision-making process
v. Improvement in employees' satisfaction
vi. Reduction of defects and rejects

1.8 Strategies for Acquiring Computer/ICT/Computing Resources

There are several options for acquiring ICT resources. Some of them are:

i. Buy: Buy the ICT resources imply that the organisation has to put up a high initial capital investment in order to pay for these resources. Apart from that, these resources need to be maintained. An advantage of the buy option is that these resources can be considered as assets and hence can increase the value of the organisation.

ii. Lease: Instead of buying the ICT resources, organizations can lease these resources from ICT leasing companies. Compared with the buy option, the lease option can result in substantial cost saving. Furthermore, most of leasing agreements include maintenance and this can reduce a lot of overhead especially for smaller organizations.

iii. Outsource: The third option is to outsource the ICT services. In this approach, all ICT services required by an organisation will be maintained and operated by the outsourcing company.

In the case of software, there is another option available, that is, to develop the software.

1.9 Importance of computer systems

Information has been considered the most important asset of an organisation or a

society. Without a proper and sufficient amount of information, an organisation or a society will not be able to make the right decision. Computer systems enable an organisation or a society to access large amount of information and/or data. Also, computer systems enable us to communicate and transfer information from one place to another at a very rapid rate. The development of computer systems or advancement in computing has introduced four powerful worldwide changes that have altered the business environment. Organisations or societies that do not respond to these changes will not be able to survive and prosper in the new business environment. These changes are listed in the Table below.

New environment	**Changes to business environment**
Globalisation	Global marketplace
	Global markets
	Global workgroups
	Global delivery systems
	Borderless world
Knowledge economy	New product and services
	Time-based competition
	Shorter product life cycle
Transformation of business enterprises	Flatter organisations
	Decentralisation
	Employees' flexibility
	Location independence
	Empowerment
	Collaborative work and teamwork
Emergence of the digital organisations	Extensive use of digital network
	Digital relationship with customers and suppliers
	Digital management
	Rapid sensing and responding to environmental changes

Globalisation has created a global marketplace and a global competition between organisations. The success of any business organisation today and in the future depends on the ability to operate in this global marketplace. ICT provides the ability for an organisation to manage and use information, as well as to communicate with various parties - customers, suppliers, and distributors all over the world, 24 hours a day. ICT also enables business organisations to buy and sell their products online. Thus, electronic commerce (e-commerce) allows trading to be done more efficiently.

Business in this contemporary world operates in knowledge economy or digital economy. In this economy, knowledge and information intense products and

services are becoming key ingredients in creating wealth. In developed countries, for example the United States of America, knowledge and information intense products and services now contribute 60% of the Gross Domestic Product (GDP) and employ nearly 55% of the labour force. In order to be successful in this knowledge economy, business organisations must be able to optimise the flow of information and knowledge within the organisation as well as to maximise the organisations knowledge resources. Ghana is experiencing this knowledge economy via digitalisation who most of our workforces are controlled by digital technologies.

The current business environment has changed a traditional business organisation from a hierarchical, centralised, and structured arrangement. This strategy has flattened and decentralised organisational activities with an easy access of information. The traditional management group relies on formal plans, a rigid division of labour and formal rules. The new management style relies on informal commitments and network to establish goals, a flexible arrangement of teams working in task forces, and a customer orientation, in order to achieve coordination between employees. In order for this style of management to be successful, it has to be supported by sufficient ICT infrastructure.

The intensive use of computers, ICT or computing in business organisations has created the concept of digital organisations. A digital organisation can be defined as an organisation where nearly all of the organisation's significant business activities are digitally enabled and mediated. In a digital organisation, any piece of information required to support key business decisions is available at anytime and anywhere in the organisation.

In the years between 2010 and 2020, I stated that the computer systems, computing or ICT development in Ghana will focus on developing Ghana as a knowledge society and we have seen the outcome. A knowledge society is a society in which knowledge becomes major creative force. The establishment of Ghana as a knowledge society is very important in order to enable us to be a leader in the knowledge economy. Knowledge economy is where a country's workforce is controlled by digital technology.